



Ratio Tables That Tell a Story Teaching Bundle

Grade 6 • North Carolina • Mathematics

NC.6.RP.3 • Use unit rates and equivalent ratios.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Use unit rates and equivalent ratios. This bundle practices a focused part of NC.6.RP.3; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Find the amount per one unit. Multiply both quantities by the same scale factor and label units.

Example 1: The science club uses 15 labels for 3 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

Unit rate: $15 \div 3 = 5$ labels/kit. Table pairs: (1,5), (3,15), (11,55).

Example 2: The design team uses 4 labels for 2 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

Unit rate: $4 \div 2 = 2$ labels/kit. Table pairs: (1,2), (2,4), (12,24).

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

The reading group uses 16 labels for 8 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.

Explain your method.

CARD 02

The music club uses 35 labels for 7 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

Explain your method.

CARD 03

The science club uses 20 labels for 5 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

Explain your method.

CARD 04

The music club uses 63 labels for 9 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

The library team uses 6 labels for 3 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

Explain your method.

CARD 06

The music club uses 12 labels for 2 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

Explain your method.

CARD 07

The community center uses 24 labels for 8 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

Explain your method.

CARD 08

The community center uses 35 labels for 5 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

Explain your method.



Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

The design team uses 24 labels for 4 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

Explain your method.

CARD 10

The school garden uses 49 labels for 7 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

Explain your method.

CARD 11

The science club uses 24 labels for 6 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

Explain your method.

CARD 12

The music club uses 15 labels for 5 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

The design team uses 4 labels for 2 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

Explain your method.

CARD 14

The reading group uses 8 labels for 2 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

Explain your method.

CARD 15

The art club uses 32 labels for 8 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

Explain your method.

CARD 16

The music club uses 24 labels for 8 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

Explain your method.



Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. The library team uses 40 labels for 8 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

2. The school garden uses 25 labels for 5 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. The reading group uses 72 labels for 9 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

4. The art club uses 35 labels for 5 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. The design team uses 9 labels for 3 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

6. The reading group uses 28 labels for 4 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. The reading group uses 21 labels for 3 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

8. The library team uses 64 labels for 8 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. The school garden uses 48 labels for 8 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.

10. The art club uses 9 labels for 3 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The science club uses 27 labels for 9 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

2. The school garden uses 35 labels for 5 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

3. The reading group uses 49 labels for 7 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. The community center uses 27 labels for 9 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

5. The library team uses 4 labels for 2 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

6. The reading group uses 32 labels for 8 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. The design team uses 48 labels for 8 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

2. The reading group uses 12 labels for 3 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

3. The science club uses 21 labels for 3 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

4. The design team uses 40 labels for 5 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.



Task-card worked solutions

1. 20 labels

Unit rate: $16 \div 8 = 2$ labels/kit. Table pairs: (1,2), (8,16), (10,20).

2. 55 labels

Unit rate: $35 \div 7 = 5$ labels/kit. Table pairs: (1,5), (7,35), (11,55).

3. 44 labels

Unit rate: $20 \div 5 = 4$ labels/kit. Table pairs: (1,4), (5,20), (11,44).

4. 63 labels

Unit rate: $63 \div 9 = 7$ labels/kit. Table pairs: (1,7), (9,63), (9,63).

5. 18 labels

Unit rate: $6 \div 3 = 2$ labels/kit. Table pairs: (1,2), (3,6), (9,18).

6. 72 labels

Unit rate: $12 \div 2 = 6$ labels/kit. Table pairs: (1,6), (2,12), (12,72).

Task-card worked solutions

7. 21 labels

Unit rate: $24 \div 8 = 3$ labels/kit. Table pairs: (1,3), (8,24), (7,21).

8. 56 labels

Unit rate: $35 \div 5 = 7$ labels/kit. Table pairs: (1,7), (5,35), (8,56).

9. 72 labels

Unit rate: $24 \div 4 = 6$ labels/kit. Table pairs: (1,6), (4,24), (12,72).

10. 84 labels

Unit rate: $49 \div 7 = 7$ labels/kit. Table pairs: (1,7), (7,49), (12,84).

11. 28 labels

Unit rate: $24 \div 6 = 4$ labels/kit. Table pairs: (1,4), (6,24), (7,28).

12. 33 labels

Unit rate: $15 \div 5 = 3$ labels/kit. Table pairs: (1,3), (5,15), (11,33).

Task-card worked solutions

13. 16 labels

Unit rate: $4 \div 2 = 2$ labels/kit. Table pairs: (1,2), (2,4), (8,16).

14. 36 labels

Unit rate: $8 \div 2 = 4$ labels/kit. Table pairs: (1,4), (2,8), (9,36).

15. 44 labels

Unit rate: $32 \div 8 = 4$ labels/kit. Table pairs: (1,4), (8,32), (11,44).

16. 36 labels

Unit rate: $24 \div 8 = 3$ labels/kit. Table pairs: (1,3), (8,24), (12,36).

Warm-up worked solutions

1. 45 labels

Unit rate: $40 \div 8 = 5$ labels/kit. Table pairs: (1,5), (8,40), (9,45).

2. 50 labels

Unit rate: $25 \div 5 = 5$ labels/kit. Table pairs: (1,5), (5,25), (10,50).

3. 64 labels

Unit rate: $72 \div 9 = 8$ labels/kit. Table pairs: (1,8), (9,72), (8,64).

4. 84 labels

Unit rate: $35 \div 5 = 7$ labels/kit. Table pairs: (1,7), (5,35), (12,84).

5. 36 labels

Unit rate: $9 \div 3 = 3$ labels/kit. Table pairs: (1,3), (3,9), (12,36).

6. 84 labels

Unit rate: $28 \div 4 = 7$ labels/kit. Table pairs: (1,7), (4,28), (12,84).

Warm-up worked solutions

7. 56 labels

Unit rate: $21 \div 3 = 7$ labels/kit. Table pairs: (1,7), (3,21), (8,56).

8. 96 labels

Unit rate: $64 \div 8 = 8$ labels/kit. Table pairs: (1,8), (8,64), (12,96).

9. 60 labels

Unit rate: $48 \div 8 = 6$ labels/kit. Table pairs: (1,6), (8,48), (10,60).

10. 30 labels

Unit rate: $9 \div 3 = 3$ labels/kit. Table pairs: (1,3), (3,9), (10,30).

Classwork worked solutions

1. 36 labels

Unit rate: $27 \div 9 = 3$ labels/kit. Table pairs: (1,3), (9,27), (12,36).

2. 49 labels

Unit rate: $35 \div 5 = 7$ labels/kit. Table pairs: (1,7), (5,35), (7,49).

3. 77 labels

Unit rate: $49 \div 7 = 7$ labels/kit. Table pairs: (1,7), (7,49), (11,77).

4. 33 labels

Unit rate: $27 \div 9 = 3$ labels/kit. Table pairs: (1,3), (9,27), (11,33).

5. 16 labels

Unit rate: $4 \div 2 = 2$ labels/kit. Table pairs: (1,2), (2,4), (8,16).

6. 48 labels

Unit rate: $32 \div 8 = 4$ labels/kit. Table pairs: (1,4), (8,32), (12,48).

Exit-ticket worked solutions

1. 54 labels

Unit rate: $48 \div 8 = 6$ labels/kit. Table pairs: (1,6), (8,48), (9,54).

2. 32 labels

Unit rate: $12 \div 3 = 4$ labels/kit. Table pairs: (1,4), (3,12), (8,32).

3. 84 labels

Unit rate: $21 \div 3 = 7$ labels/kit. Table pairs: (1,7), (3,21), (12,84).

4. 56 labels

Unit rate: $40 \div 5 = 8$ labels/kit. Table pairs: (1,8), (5,40), (7,56).

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The October book drive uses 35 labels for 5 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

2. The leaf observation team uses 49 labels for 7 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

3. The September art display uses 32 labels for 4 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.

4. The October book drive uses 10 labels for 5 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The October book drive uses 9 labels for 3 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.

6. The fall garden club uses 12 labels for 6 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

7. The leaf observation team uses 27 labels for 9 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

8. The September art display uses 16 labels for 2 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.



Fall • Worked solutions

1. 77 labels

Unit rate: $35 \div 5 = 7$ labels/kit. Table pairs: (1,7), (5,35), (11,77).

2. 77 labels

Unit rate: $49 \div 7 = 7$ labels/kit. Table pairs: (1,7), (7,49), (11,77).

3. 80 labels

Unit rate: $32 \div 4 = 8$ labels/kit. Table pairs: (1,8), (4,32), (10,80).

4. 16 labels

Unit rate: $10 \div 5 = 2$ labels/kit. Table pairs: (1,2), (5,10), (8,16).

Fall • Worked solutions

5. 30 labels

Unit rate: $9 \div 3 = 3$ labels/kit. Table pairs: (1,3), (3,9), (10,30).

6. 24 labels

Unit rate: $12 \div 6 = 2$ labels/kit. Table pairs: (1,2), (6,12), (12,24).

7. 33 labels

Unit rate: $27 \div 9 = 3$ labels/kit. Table pairs: (1,3), (9,27), (11,33).

8. 80 labels

Unit rate: $16 \div 2 = 8$ labels/kit. Table pairs: (1,8), (2,16), (10,80).

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The December art display uses 12 labels for 4 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

2. The January puzzle club uses 21 labels for 3 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

3. The winter supply drive uses 20 labels for 5 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

4. The December art display uses 25 labels for 5 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The winter reading club uses 42 labels for 7 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

6. The December art display uses 12 labels for 6 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

7. The indoor garden project uses 27 labels for 9 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

8. The winter reading club uses 32 labels for 4 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.



Winter • Worked solutions

1. 21 labels

Unit rate: $12 \div 4 = 3$ labels/kit. Table pairs: (1,3), (4,12), (7,21).

2. 49 labels

Unit rate: $21 \div 3 = 7$ labels/kit. Table pairs: (1,7), (3,21), (7,49).

3. 36 labels

Unit rate: $20 \div 5 = 4$ labels/kit. Table pairs: (1,4), (5,20), (9,36).

4. 55 labels

Unit rate: $25 \div 5 = 5$ labels/kit. Table pairs: (1,5), (5,25), (11,55).

Winter • Worked solutions

5. 48 labels

Unit rate: $42 \div 7 = 6$ labels/kit. Table pairs: (1,6), (7,42), (8,48).

6. 22 labels

Unit rate: $12 \div 6 = 2$ labels/kit. Table pairs: (1,2), (6,12), (11,22).

7. 24 labels

Unit rate: $27 \div 9 = 3$ labels/kit. Table pairs: (1,3), (9,27), (8,24).

8. 80 labels

Unit rate: $32 \div 4 = 8$ labels/kit. Table pairs: (1,8), (4,32), (10,80).

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The March art display uses 14 labels for 2 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

2. The spring book swap uses 63 labels for 9 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

3. The April science team uses 24 labels for 4 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.

4. The spring reading fair uses 18 labels for 6 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The April science team uses 4 labels for 2 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

6. The spring book swap uses 42 labels for 6 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

7. The spring reading fair uses 42 labels for 7 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.

8. The spring book swap uses 15 labels for 3 equal kits. At the same rate, how many labels are needed for 11 kits? Make a ratio table including one kit.



Spring break • Worked solutions

1. 77 labels

Unit rate: $14 \div 2 = 7$ labels/kit. Table pairs: (1,7), (2,14), (11,77).

2. 49 labels

Unit rate: $63 \div 9 = 7$ labels/kit. Table pairs: (1,7), (9,63), (7,49).

3. 60 labels

Unit rate: $24 \div 4 = 6$ labels/kit. Table pairs: (1,6), (4,24), (10,60).

4. 36 labels

Unit rate: $18 \div 6 = 3$ labels/kit. Table pairs: (1,3), (6,18), (12,36).

Spring break • Worked solutions

5. 14 labels

Unit rate: $4 \div 2 = 2$ labels/kit. Table pairs: (1,2), (2,4), (7,14).

6. 56 labels

Unit rate: $42 \div 6 = 7$ labels/kit. Table pairs: (1,7), (6,42), (8,56).

7. 66 labels

Unit rate: $42 \div 7 = 6$ labels/kit. Table pairs: (1,6), (7,42), (11,66).

8. 55 labels

Unit rate: $15 \div 3 = 5$ labels/kit. Table pairs: (1,5), (3,15), (11,55).

Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The June art workshop uses 40 labels for 5 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.

2. The summer library club uses 54 labels for 9 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

3. The summer supply team uses 10 labels for 2 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

4. The July puzzle camp uses 16 labels for 2 equal kits. At the same rate, how many labels are needed for 10 kits? Make a ratio table including one kit.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The community garden uses 42 labels for 6 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

6. The summer library club uses 12 labels for 4 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

7. The summer supply team uses 12 labels for 2 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

8. The summer supply team uses 14 labels for 2 equal kits. At the same rate, how many labels are needed for 9 kits? Make a ratio table including one kit.



Summer • Worked solutions

1. 72 labels

Unit rate: $40 \div 5 = 8$ labels/kit. Table pairs: (1,8), (5,40), (9,72).

2. 42 labels

Unit rate: $54 \div 9 = 6$ labels/kit. Table pairs: (1,6), (9,54), (7,42).

3. 35 labels

Unit rate: $10 \div 2 = 5$ labels/kit. Table pairs: (1,5), (2,10), (7,35).

4. 80 labels

Unit rate: $16 \div 2 = 8$ labels/kit. Table pairs: (1,8), (2,16), (10,80).



Summer • Worked solutions

5. 49 labels

Unit rate: $42 \div 6 = 7$ labels/kit. Table pairs: (1,7), (6,42), (7,49).

6. 36 labels

Unit rate: $12 \div 4 = 3$ labels/kit. Table pairs: (1,3), (4,12), (12,36).

7. 48 labels

Unit rate: $12 \div 2 = 6$ labels/kit. Table pairs: (1,6), (2,12), (8,48).

8. 63 labels

Unit rate: $14 \div 2 = 7$ labels/kit. Table pairs: (1,7), (2,14), (9,63).

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use unit rates and equivalent ratios. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Find the amount per one unit. Multiply both quantities by the same scale factor and label units.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The school art team uses 8 labels for 4 equal kits. At the same rate, how many labels are needed for 7 kits? Make a ratio table including one kit.

2. The student design group uses 28 labels for 4 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

3. The classroom supply team uses 36 labels for 9 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

4. The school art team uses 49 labels for 7 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.



Substitute lesson • Worked solutions

1. 14 labels

Unit rate: $8 \div 4 = 2$ labels/kit. Table pairs: (1,2), (4,8), (7,14).

2. 56 labels

Unit rate: $28 \div 4 = 7$ labels/kit. Table pairs: (1,7), (4,28), (8,56).

3. 48 labels

Unit rate: $36 \div 9 = 4$ labels/kit. Table pairs: (1,4), (9,36), (12,48).

4. 56 labels

Unit rate: $49 \div 7 = 7$ labels/kit. Table pairs: (1,7), (7,49), (8,56).

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use unit rates and equivalent ratios. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Find the amount per one unit. Multiply both quantities by the same scale factor and label units.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The desk-supply team uses 15 labels for 5 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

2. The paper puzzle team uses 18 labels for 9 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.

3. The paper puzzle team uses 64 labels for 8 equal kits. At the same rate, how many labels are needed for 12 kits? Make a ratio table including one kit.

4. The classroom book group uses 56 labels for 8 equal kits. At the same rate, how many labels are needed for 8 kits? Make a ratio table including one kit.



Emergency substitute lesson • Worked solutions

1. 36 labels

Unit rate: $15 \div 5 = 3$ labels/kit. Table pairs: (1,3), (5,15), (12,36).

2. 16 labels

Unit rate: $18 \div 9 = 2$ labels/kit. Table pairs: (1,2), (9,18), (8,16).

3. 96 labels

Unit rate: $64 \div 8 = 8$ labels/kit. Table pairs: (1,8), (8,64), (12,96).

4. 56 labels

Unit rate: $56 \div 8 = 7$ labels/kit. Table pairs: (1,7), (8,56), (8,56).

A completed lesson-plan example

Level and focus

Grade 6; NC.6.RP.3; Use unit rates and equivalent ratios.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.6.RP.3 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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